

TourAtlas

Tourism that works offline · Tourism Case C · Hossam Elshahaby

Overview

TourAtlas is an offline-first, mobile-first workspace for local tourism operators in rural, cultural, eco-trail, youth-led and conflict-affected settings. A visitor speaks or writes in their own language; the operator understands the request, approves a reply in their own words, and the app keeps working when connectivity disappears.

The problem

- Signal is weakest exactly where rural and heritage tourism happens.
- Visitors and operators rarely share a language.
- Knowledge about historical places is fragile during crises and conflict.

Key features

- **Multi-country packs** — Gambia, Jordan, Ukraine and any operator-created region, each with isolated records.
- **Voice in and out** — hold-to-speak, online transcription, on-device Whisper offline, spoken replies.
- **Translation with human approval** — online AI or on-device OPUS-MT; nothing is sent without the operator.
- **Smart conversations** — dates, times, group size, budget and meeting points accumulate across turns.
- **Small on-device AI** — a compact WebGPU model drafts replies from the operator's own facts.
- **Offline maps** — 3D explorer plus prepared and custom map packs that reopen with the network off.
- **SMS fallback** — approved replies via SMS or the phone's messaging app.
- **Heritage protection** — private notes on historical places, synced only on explicit opt-in.
- **Feedback insights & evidence** — reviews analysed locally; every capability honestly labelled.

Impact

- More bookings and income for small, rural and youth-led operators.
- Reliable communication without coverage.
- Protection of cultural heritage during crisis.
- Transparent, measured evidence instead of unverified claims.

Technology

TanStack Start (React 19), Tailwind CSS, Leaflet and WebGL maps, transformers.js and web-llm for on-device models, browser-local storage with opt-in cloud sync, AI gateway for online translation and speech, Twilio for SMS.

Links

Live app	travelcase-explorer.lovable.app
Team introduction	https://youtu.be/oFYcHAJsyc
Product demo	https://youtu.be/vU0QsG1MOB4
Technical depth	https://youtu.be/CNfpmA17Ng4